

Valuation Process and Cost of Production Report

A number of stages are passed through in the valuation necessary in process costing for cost statement.

- First, the physical flow of the units of production must be calculated having regards to the total number of units to be accounted for, regardless of the degree of completion.

- Secondly, the equivalent units involved in the physical flow are to be calculated.

- Thirdly, the total equivalent units and the current equivalent units involved are to be calculated. These are to be accounted for in respect of the cost elements (transferred-in cost, material cost and conversion cost).

- Fourthly, the unit costs are to be calculated, paying regards to the stock valuation method assumed (FIFO, WAP (Weighted Average Price), LIFO, etc.).

- Finally, the total cost of the transferred-out products and work-in-process are to be calculated, ensuring that all costs are accounted for.

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Cost of Production Report

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Illustration 1

Process 2 receives units from process 1 and after carrying out work on the units transfers them to process 3. For one accounting period the relevant data for process 2 were as follows:

Opening WIP 200 units (25% completed) valued at #2500

800 units received from process 1 valued at #4300

840 units were transferred to process 3

Closing WIP 160 units (50% completed)

The costs for the period were #16580 and no units were scrapped. It is required to prepare the process accounts for Process 2 using

1. FIFO method of valuation
2. The average cost method of valuation

Illustration 2

The following data relates to Process Y for accounting period 2.

At the beginning of Period 2 there were 800 units partly completed which had the following values

	#	% agc
completed		
Input material (from Process X)	8200	100
Material introduced	5600	55
Labour	3200	60
Overheads	2400	45

During the periods 4300 units were transferred from Process X at a value of #46,500 and other cost were:

Material introduced	#24,000
Labour	#19500
Overheads	#18200

At the end of the period, the closing WIP was 600 units which were at the following stage of completion:

Input material	100% complete
Material introduced	50% complete
Labour	45% complete

Overheads

*40% complete

The balance of 4500 units was transformed to finished goods

Required: Calculate the value of units transferred to finished goods and the value of WIP and prepare the process account using FIFO and Average Cost method

Joint Products and By-Products

The Concept of Joint Products

The process costing principles discussed in Paragraph 7.1 are shown to be all about determining the cost of processing some inputs that yield the same types of products. At the end of the processing activities, only one type of product would result from the processed raw material.

However, it is not always that we have only one type of product from a processing operation. It is possible for a single raw material to yield two or more products simultaneously when processed. Such products are known as Joint Products. For example, when crude oil (a single raw material) is processed or refined, petrol, kerosene, gas, etc., could be obtained from it.

The cost of processing a production input (raw material) that would amount to joint products is known as Joint Cost. The joint cost is to be restricted to the split-off point (point after which each joint product would be incurring separate processing cost). Joint cost is not to be traced to any particular product but rather to all the joint products as a group. There are many ways of apportioning joint cost to joint products for financial accounting purposes.

In practice, it is normal to identify one product out of the joint products as the main or principal product and the rest to be treated as joint products or even by-products. In the example above, it is clear that petrol is the main product to be identified as crude oil is processed. Pairs of shoes could be identified as crude oil is processed, while bags, wallets, etc., could be seen as joint or by-products.

One way of differentiating between by-product and joint product is to consider their cost of production or sales value. A product that costs 10% to 15% of the main product cost should be treated as a by-product. Any product that costs 15% to 40% of the main product cost is a joint product. Any product that costs above 40% of the identified main product cost should also be treated as a main product. As a result of changes in price, therefore, a by-product can become a joint-product or even a main product and vice versa.

Accounting Treatment of By- Products

A by-product is a secondary product arising as a result of a processing activity aimed at producing a certain main product. The market value of a by-product less the processing cost after the split off point is usually negligible, compared to the total market value of all the joint products or the market value of the main product.

The usual treatment of by-product is to deduct its Net Realizable Value (NRV) from the total Joint Cost (JC) and then divide the net joint cost among the joint or main products. The NRV of the by-product is the difference between its market value and its separate processing cost.

Illustration 1 Treatment of by- products

Wambai Shoemakers have a process that yields two main products: A and B and a by-product C at a total cost of #3,000,000. There are 1,000 units of C requiring no further processing and each can be sold at #60 with negligible market cost. The two main products take equal share of joint cost.

Required: What should be the share of Product A from the Joint Cost?

Methods of Apportionment of Joint Cost

There are three popular methods of sharing joint cost to the joint (or main) products. These are the Physical Units Bases, Sales Value (at the point of separation) and Net Realization Basis. These methods are briefly discussed below:

(a) Physical Units Basis

Under this method, the joint cost is shared among the joint products on the basis of the quantities of physical units, provided all the products are measured by a common unit of measurement, such as kilograms or litres. The problem with this method is that

consideration is not given to price and, so, it does not consider the value of the products. Usually, the value of products is the most important factor to be considered.

Illustration 2

Anadarya Company Ltd., Tiga has a processing system that produces three products: K, S and T with 5,000 kg, 3,000 kg and 2,000 kg, respectively, in a year. The total cost incurred up to the split-off point in the year 2000 was N1,000,000. Use the physical units basis to share the joint cost among the three products. Calculate also their unit cost.

(b) Sales Value (at the Point of Separation)

Under this method, the joint cost is shared among the joint products on the basis of their sales value before further processing. At the split off point, market value can be estimated per unit of each of the joint products. It is the ratios of the sales value of the joint products that are to be used as basis of apportioning the joint cost.

The problems with this method are two-fold: One, a product may have zero value at the point of separation but significant value with little processing cost after the split-off point. Secondly, a product may have high selling price at the split-off point and hence high sales value but may involve large selling and distribution cost (advert, carriage, etc.).

Illustration 3

Assuming that Anadarya Company Ltd has estimated the following selling prices for its three products at the point of separation:

$$K = \text{N}400/\text{unit}$$

$$S = \text{N}440/\text{unit}$$

$$T = \text{N}340/\text{unit}$$

Use the Sales Value method to apportion the joint cost and determine the unit cost of each of the three products.